Endometriosis Longitudinal Intelligence Platform

Product and Technical Design

1. Product Definition

This is not a period tracker and it is not an AI diagnostic chatbot.

It is a **longitudinal health-intelligence platform** designed around a simple problem:

> Important endometriosis-related information develops across days, menstrual cycles, treatments and years, while a clinical consultation sees only a short snapshot.

The platform continuously builds a structured history of:

Symptoms → menstrual context → functional impact → medication → treatment → investigations → healthcare events

It then calculates meaningful patterns from that history and uses AI to turn those patterns into a concise, evidence-backed clinical summary.

```
PATIENT EXPERIENCE
        │
        ▼
Low-burden daily capture
        │
        ▼
Structured longitudinal record
        │
        ▼
Deterministic pattern analysis
        │
        ▼
AI explanation and summarization
        │
        ▼
Patient review and selective sharing
        │
        ▼
CLINICIAN
```

The AI does not diagnose endometriosis.

Its role is to make the patient's longitudinal history **visible, interpretable and clinically usable**.

---

2. Core Design Principles

Every design decision has to satisfy four requirements.

Requirement	Design response
Patient keeps using the app	Daily interaction remains extremely short
Information remains clinically useful	Symptoms include timing, context and impact
AI remains trustworthy	Calculations happen outside the LLM
Patient feels safe	Data minimization and patient-controlled sharing

The objective is therefore not to collect the maximum possible information every day.

It is to collect the **minimum information required at the right time**.

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3. What We Need to Understand About Each Patient

The full longitudinal model ultimately needs to understand seven things:

1. What symptoms occur?

2. When do they occur?

3. How severe are they?

4. How much do they affect everyday life?

5. What other symptoms occur with them?

6. What medication or treatment was active?

7. How has the pattern changed over time?

The system collects this progressively rather than through one huge questionnaire.

The literature supporting the project specifically identifies the need to capture pain alongside gastrointestinal, urinary, fatigue, emotional and functional domains while keeping daily tracking low burden.

---

4. Complete Patient Data Model

Patient information is divided into:

```
BASELINE DATA
Collected occasionally

+

DAILY DATA
Collected quickly

+

EVENT-BASED DATA
Asked only when something happens

+

DERIVED DATA
Calculated automatically
```

This distinction is central to usability.

---

5. Baseline Patient Profile

These questions are collected during onboarding or gradually over the first few sessions.

They are not asked daily.

General Clinical Context

Information	How collected
Age / age range	Once
Suspected or confirmed endometriosis	Once/update

Information	How collected
Approximate symptom onset	Once
Family history of endometriosis	Once
Relevant previous diagnoses	Once/update
Major existing health conditions	Optional
Previous pelvic surgery	Once
Pregnancy history	Optional
Fertility concerns	Optional
Current reproductive goals	Optional

Exact DOB, address, Aadhaar and unnecessary identifying information are not required by the health-intelligence engine.

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6. Menstrual and Reproductive Baseline

The system records:

Variable	Example
Age at menarche	12
Usual cycle length	27–32 days
Cycle regularity	regular / variable / highly irregular
Typical bleeding duration	5 days
Heavy menstrual bleeding	yes/no/unsure
Spotting	yes/no
Last menstrual period	date
Hormonal contraception	current/past
Amenorrhoea due to treatment	yes/no
Pregnancy possibility	optional
Fertility treatment	optional

The cycle engine does not assume a standard 28-day cycle.

It learns from actual recorded bleeding.

7. Core Symptom Domains

The application should support all clinically relevant symptom groups.

A. Pelvic and Abdominal Pain

Track:

- pelvic pain;
- lower abdominal pain;
- menstrual pain;
- pain before menstruation;
- pain during menstruation;
- pain after menstruation;
- pain outside menstruation;
- ovulation-associated pain where reported;
- lower-back pain;
- hip pain;
- leg pain;
- rectal pressure/pain;
- groin pain.

For a pain episode, optional detail includes:

```
Severity
Location
Duration
Character
Trigger
Functional impact
Medication used
```

Pain character can include:

cramping / stabbing / burning / aching / pulling / pressure / throbbing / patient-described

8. Bowel and Gastrointestinal Symptoms

Track:

Symptom	Possible detail
Painful bowel movement	severity
Constipation	duration
Diarrhoea	frequency
Bloating	severity
Abdominal distension	severity
Abdominal cramps	severity
Nausea	severity
Vomiting	event
Rectal pain	severity
Rectal bleeding	event
Change in bowel pattern	event
Urgency	yes/no

Timing is particularly important.

The system should detect whether these symptoms concentrate around menstruation or occur independently.

---

9. Urinary Symptoms

Track:

- pain during urination;
- bladder pain;
- urinary urgency;
- urinary frequency;
- difficulty emptying bladder;
- cyclical urinary discomfort;
- visible blood in urine;
- pelvic pressure associated with urination.

Rare or concerning findings are not used by AI to diagnose endometriosis.

They can instead trigger appropriate general medical safety messaging.

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10. Sexual Symptoms

This entire module is optional.

The system can track:

- pain during intercourse;
- deep pelvic pain;
- pain after intercourse;
- duration of post-intercourse pain;
- avoidance because of anticipated pain;
- effect on intimate activity.

The app should never repeatedly ask:

> Did you have intercourse today?

Instead, this module appears only if the patient enables it.

The user can manually record an event when relevant.

---

11. Bleeding

Daily bleeding is simple:

```
None
Spotting
Light
Moderate
Heavy
Very heavy / unusual
```

Optional event detail can include:

- clots;
- breakthrough bleeding;
- bleeding after intercourse;
- unexpected bleeding;
- perceived change from normal.

The purpose is not just menstrual tracking.

Bleeding provides temporal anchors against which other symptoms can be compared.

---

12. Fatigue and Systemic Symptoms

Track:

- fatigue;
- exhaustion;
- reduced energy;
- dizziness;
- nausea;
- headache/migraine;
- sleep disturbance;
- generalized discomfort;
- patient-defined recurring systemic symptoms.

Fatigue should be tracked separately from pain.

A patient can have moderate pain but severe fatigue-related functional impairment.

---

13. Rare or Extrapelvic Symptoms

These should exist in the system without cluttering daily use.

Examples include:

- cyclical shoulder pain;
- cyclical chest pain;
- breathing symptoms associated with menstruation;
- cyclical cough;
- scar pain or swelling;
- cyclical pain at unusual body locations.

These are available through:

Add another symptom

rather than appearing on everyone's home screen.

---

14. Functional Impact

This is one of the most important elements in the system.

The app should separate:

```
SYMPTOM SEVERITY

from

LIFE IMPACT
```

Because:

```
Pain 8/10 + normal day

is different from

Pain 8/10 + unable to attend work
```

The literature specifically supports measuring functional impact rather than treating pain intensity as the complete disease burden.

Impact Domains

The system can record whether symptoms affected:

Area	Example
Work	missed work/reduced productivity
Education	missed class/unable to study
Exercise	reduced/avoided
Walking/mobility	difficulty
Sleep	disrupted
Household activities	unable to complete
Social activity	cancelled plans
Sexual activity	reduced/avoided
Concentration	impaired
Emotional wellbeing	affected

However, the patient is not expected to fill all of these daily.

The standard daily question is simply:

Did your symptoms interfere with your normal day?

```
No
A little
I had to modify activities
I could not perform an important activity
```

Only if necessary does the application ask:

What was affected?

This gives us rich impact information without daily questionnaire burden.

---

15. Medication Tracking

The system maintains the patient's regular medication list.

The patient does not type medication names every day.

Once configured, daily logging becomes:

```
Medication today?

[Ibuprofen]
[Paracetamol]
[Other]
[None]
```

For medication events we can capture:

```
Medication
Dose
Time
Reason
Pain before medication
Pain after medication
Perceived effectiveness
Side effects
```

But detailed before/after response is requested only occasionally or during significant episodes.

This lets the system study:

symptom → medication → subsequent symptom behaviour

without turning the app into a medication diary.

---

16. Treatment Episodes

Longer-term interventions are stored differently from daily medicines.

Examples include:

- oral contraceptive;
- progesterone therapy;
- GnRH therapy;
- IUD;
- physiotherapy;
- pain-management intervention;
- surgery;
- fertility treatment;
- other clinician-directed therapy.

Each treatment episode stores:

```
Treatment
Start date
Stop date
Dose where relevant
Reason
Side effects
Patient-perceived change
```

The analytical engine then compares periods:

```
BEFORE TREATMENT

vs

DURING TREATMENT

vs
```

```
AFTER TREATMENT
```

for multiple symptom domains.

---

17. Healthcare Journey

The system also tracks important clinical events.

These include:

- GP consultation;
- gynecologist consultation;
- emergency visit;
- previous diagnosis;
- referral;
- ultrasound;
- MRI;
- CT;
- laparoscopy;
- surgery;
- fertility consultation;
- treatment recommendation;
- hospital admission.

An event can store:

```
Date
Type
Reason
Outcome
Document attachment
Patient note
```

This allows the system to reconstruct:

```
Symptoms
   ↓
Consultation
   ↓
Investigation
   ↓
Treatment
```

```
   ↓
Symptoms continue
   ↓
Another consultation
```

rather than viewing each event independently.

---

18. Patient-Defined Symptoms

A fixed symptom checklist is insufficient because endometriosis presentations vary substantially between patients.

The application therefore always has:

+ Add my own symptom

The patient might enter:

> “Sharp pain going down my left thigh.”

AI can suggest:

```
Would you like to save this as:

Left-leg pain
Sharp
```

The patient confirms it.

From then onward:

Left-leg pain

can appear as one of that person's quick-access symptoms.

The patient's own terminology is retained alongside the standardized internal representation.

---

19. What Is Actually Asked Every Day?

This is the key usability question.

The complete data model may contain more than one hundred possible fields.

The patient should normally see only four or five questions.

A routine day could be:

```
TODAY

Bleeding?
○ None ○ Spotting ○ Light ○ Moderate ○ Heavy

Pelvic pain?
0 ———————————— 10

Anything else?
[Bowel pain] [Fatigue] [Back pain] [+ More]

Did symptoms affect your day?
○ No
○ A little
○ Had to change my activities
○ Could not perform an important activity

Medication?
○ None
[Ibuprofen]
[Other]
```

Done.

The rest of the database is built from conditional, historical and automatically derived information.

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20. The App Should Learn When Not to Ask

Personalization is not simply:

> recommending content.

The more important form of personalization is **question reduction**.

Suppose a patient has never experienced urinary symptoms after three months.

The app should not ask about urinary symptoms every day.

Suppose another patient reports painful bowel movements almost every menstruation.

During the relevant cycle window, the app might surface:

> Any bowel pain today?

This produces an adaptive questionnaire.

```
STATIC QUESTIONNAIRE
Same questions forever

becomes

ADAPTIVE QUESTIONNAIRE
Relevant questions at relevant times
```

---

21. Three Levels of Daily Logging

The user can choose how much detail to give.

Level 1 — Quick Check-in

Around 15–20 seconds.

```
Pain
Bleeding
Impact
Major symptom
```

Enough to preserve longitudinal continuity.

Level 2 — Standard Entry

Around 30–45 seconds.

Adds:

```
other symptoms
medication
basic impact
```

This should be the default.

Level 3 — Detailed Episode

Used only when something significant occurs.

Can include:

```
exact pain location
character
duration
trigger
bowel/urinary detail
medication response
activity affected
free-text note
```

The patient chooses the depth.

---

22. Significant Event Logging

Instead of making every day detailed, the app places detail around important events.

For example:

> Pain suddenly became severe.

The app can ask:

```
Where?
How severe?
When did it start?
What were you doing?
Any bowel/urinary symptoms?
Did it stop you from normal activity?
Medication taken?
```

That rich information is valuable because something clinically meaningful occurred.

On an ordinary day, those questions never appear.

---

23. Conversational Input

Some patients will find tapping forms tedious.

They should be able to say:

> “Had really bad cramps this morning, around 8 out of 10. Had loose stools and couldn't go to college. Took ibuprofen around ten and felt better by lunch.”

The intake agent extracts:

```
Pelvic/menstrual pain = 8
Timing = morning
Diarrhoea = present
Functional impact = missed college
Medication = ibuprofen
Response = improved
```

Then presents:

> **I understood this:**

for patient confirmation.

The AI does the data entry.

The patient provides the experience.

---

24. Logging by Exception

After the system understands the user's normal baseline, another option becomes possible.

Instead of entering everything:

> **Same as usual today?**

```
○ Yes
○ Something changed
```

If **yes**, an abbreviated entry is recorded according to explicitly confirmed baseline fields.

If something changed, the relevant questions open.

This could dramatically improve long-term retention.

It should only be introduced once the patient's baseline is established.

---

25. Smart Reminders

Reminder frequency should also adapt.

No guilt.

No streaks.

No:

> You failed to track yesterday.

Instead:

> Your check-in is ready when you need it.

Users can choose:

```
Daily
Only around menstruation
Custom schedule
No reminders
```

The system may suggest more consistent logging when data gaps prevent meaningful interpretation, but the user remains in control.

---

26. Cycle-Aware Questioning

The cycle engine can reduce unnecessary questioning further.

During menstruation, likely high-value questions may surface more prominently.

Outside menstruation, the interface can prioritize:

```
persistent pelvic pain
fatigue
non-cyclical symptoms
treatment effects
```

Importantly, the system still allows any symptom at any time.

Cycle-aware UX controls **priority**, not medical assumptions.

---

27. Personal Home Screen

After enough data are collected, each user effectively gets a different app home screen.

Patient A

```
Pelvic pain
Bowel pain
Fatigue
Back pain
```

Patient B

```
Menstrual pain
Bladder pain
Bloating
Migraine
```

Patient C

```
Pelvic pain
Leg pain
Dyspareunia
Fatigue
```

This is genuine personalization.

The system learns what is relevant without removing access to the full symptom vocabulary.

---

28. Missing Data Is Not Zero

One of the most important technical rules is:

```
NOT LOGGED ≠ NO SYMPTOM
```

The database stores three states where required:

```
Present
Absent
Unknown
```

Therefore:

```
Monday
pain = 0

Tuesday
no entry

Wednesday
pain = 7
```

Tuesday remains unknown.

The analytical engine does not infer zero.

This directly supports the literature requirement that missing-data notices and coverage be visible.

---

29. What the System Derives Automatically

The patient never manually calculates:

```
cycle day
symptom frequency
menstrual association
```

```
mean severity
median severity
peak severity
co-occurrence
functional burden
treatment response
trend
cycle-to-cycle reproducibility
```

These are derived.

For every symptom, the feature engine may calculate:

Derived feature	Meaning
Days present	basic frequency
Severe days	high-burden episodes
Median severity	typical severity
Maximum severity	worst episode
Menstrual-window rate	menstrual association
Non-menstrual rate	persistence
Recurrence	repeat across cycles
Impact days	effect on life
Medication days	treatment context
Co-occurring symptoms	symptom combinations
Trend	change over time
Coverage	reliability of observation

---

30. Meaningful Patterns

The platform is primarily looking for six classes of information.

Cyclicity

Does the symptom repeatedly appear around menstruation?

Co-occurrence

Which symptoms repeatedly appear together?

Persistence

Is a symptom becoming less episodic and more persistent?

Functional burden

Is everyday-life interference increasing even if pain intensity is stable?

Treatment response

What changed after treatment started?

Change from personal baseline

Is the patient's current experience different from her own previous pattern?

This provides clinically meaningful context without producing an artificial diagnostic score.

---

31. Cycle Analysis

The system uses actual bleeding data rather than assuming a 28-day cycle.

Symptoms are aligned relative to menstrual onset:

```
D-4
D-3
D-2
D-1
D0  ← bleeding begins
D+1
D+2
D+3
```

Multiple cycles can then be compared.

Example:

```
PAINFUL BOWEL MOVEMENTS
```

```
Cycle 1     D-1, D0, D+1
Cycle 2     D0, D+1
Cycle 3     D-2, D0, D+2
```

The feature engine can therefore establish:

> Painful bowel movements repeatedly clustered around menstruation during the observed period.

The AI does not discover this by reading raw diary entries.

The statistical engine calculates it first.

---

32. Personal Baseline

The most useful comparison is initially:

> **this patient versus herself**

rather than:

> this patient versus an average population.

Suppose:

```
Previous three cycles:
1–2 non-menstrual pain days

Latest cycle:
8 non-menstrual pain days
```

That change is potentially important even without a diagnostic classification.

The system can highlight:

> Non-menstrual pelvic-pain frequency increased relative to the patient's previous recorded cycles.

---

33. AI Architecture

The LLM comes **after** structured analytics.

```
PATIENT DATA
     |
     ▼
FEATURE ENGINE
     |
     ▼
VERIFIED FINDINGS
     |
     ▼
LLM
     |
     ▼
PLAIN-LANGUAGE SUMMARY
```

The model is never asked:

> Read the diary and tell me what is happening.

Instead it receives structured evidence.

Example:

```
Observation window: 84 days
Logged days: 71
Cycles: 3

Dyschezia:
9/11 logged menstrual-window days
5/60 other logged days

Impact:
6 affected days involved reduced normal activity
```

The LLM can then state:

> Painful bowel movements were concentrated around menstruation during the recorded period and sometimes interfered with normal activities.

Every claim remains verifiable.

---

34. AI Agents

The platform uses restricted specialized agents.

Agent	Responsibility
Intake Agent	Converts voice/text into structured candidate entries
Personalization Agent	Determines which fields should be easiest to access
Data Quality Agent	Detects incomplete or inconsistent information
Temporal Agent	Interprets calculated temporal features
Treatment Agent	Organizes treatment-related changes
Evidence Agent	Retrieves curated clinical evidence
Summary Agent	Generates understandable summaries
Fidelity Agent	Verifies summary claims against underlying data
Safety Agent	Prevents diagnostic overstatement

Agents operate inside a deterministic workflow.

They do not autonomously make clinical decisions.

---

35. Complete Data Flow

```
PATIENT
   │
   ▼
Quick daily entry / voice entry
   │
   ▼
Local encrypted storage
   │
   ▼
Schema validation
   │
   ▼
Structured event database
   │
   ├──── menstrual engine
   ├──── impact engine
   ├──── medication engine
```

```
    ├──── treatment engine
    └──── symptom engine
              │
              ▼
      Longitudinal features
              │
              ▼
       Pattern engine
              │
              ▼
     Evidence objects created
              │
              ▼
       AI summarization
              │
              ▼
       Fidelity validation
              │
              ▼
         Safety check
              │
              ▼
       Patient preview
              │
              ▼
      Selective sharing
              │
              ▼
         CLINICIAN
```

---

36. Database Design

Important entities include:

```
PatientProfile
Cycle
SymptomDefinition
PatientSymptom
SymptomEvent
ImpactEvent
BleedingEvent
MedicationEvent
TreatmentEpisode
ClinicalEvent
```

```
DerivedFeature
Consent
Report
ReportClaim
ShareGrant
AuditEvent
```

The database is event-oriented.

Health history is not continuously overwritten.

If pain changes:

```
Day 1 → 4
Day 2 → 7
Day 3 → 3
```

all three observations remain.

The change itself is meaningful information.

---

37. Technology Stack

Layer	Technology
Mobile	React Native + TypeScript
Local storage	Encrypted SQLite
Secure secrets	iOS Keychain / Android Keystore
Backend API	Python FastAPI
Validation	Pydantic
ORM	SQLAlchemy
Database	PostgreSQL
Vector retrieval	pgvector
Authentication	AWS Cognito
Documents	S3 + KMS
Async jobs	AWS SQS

Layer	Technology
Compute	AWS ECS/Fargate
Database hosting	AWS RDS
Analytics/ML	Python
AI	Provider-independent AI gateway
Infrastructure	Terraform
Monitoring	OpenTelemetry + CloudWatch
CI/CD	GitHub Actions
Containers	Docker

The stack is deliberately conventional.

For sensitive healthcare software, reliability and maintainability matter more than using fashionable infrastructure.

---

38. Offline-First Design

The patient should never lose an entry because the internet is unavailable.

```
ENTRY
  ↓
Encrypted local database
  ↓
Saved immediately
  ↓
Background synchronization
  ↓
Server acknowledgement
```

Daily logging never waits for the cloud.

This also makes the application usable in environments with inconsistent connectivity.

---

39. Privacy Architecture

Identity and health data should remain logically separated.

```
IDENTITY SYSTEM

authentication
email if needed
account recovery

        |
        | pseudonymous ID
        ▼

HEALTH SYSTEM

cycles
symptoms
impact
treatments
investigations
reports
```

The clinical intelligence layer does not need the patient's name.

---

40. Sensitive Modules Are Optional

The patient controls:

```
Menstrual tracking      ON
Pain                    ON
Bowel symptoms          ON
Urinary symptoms        ON
Sexual health           OFF
Fertility               OFF
Mental wellbeing        OFF
```

The platform remains functional even when sensitive modules are disabled.

This matters because the system contains information many users may not want stored at all.

---

41. Selective Sharing

The patient chooses what a clinician receives.

```
SHARE

✓ Pain history
✓ Cycle patterns
✓ Treatments
✓ Investigations
✓ Bowel symptoms

○ Sexual-health information
○ Fertility information
○ Private notes
```

The patient then previews the exact report.

Only the approved snapshot is shared.

Clinicians do not receive unrestricted permanent access to the patient's complete account by default.

---

42. Privacy During Everyday Use

Notifications should remain discreet.

Not:

> Record your painful intercourse symptoms.

Instead:

> Your check-in is ready.

The application should support:

```
biometric lock
PIN lock
generic notifications
hidden sensitive previews
encrypted storage
access audit
data export
data deletion
```

Privacy is therefore visible in the product rather than buried in legal documents.

43. Clinician Output

The clinician should not receive months of diary entries.

They receive a concise longitudinal summary.

Example:

Recorded Period

1 June – 31 August

Logged: 79/92 days
Cycles: 3

Main Concerns

Pelvic pain
Painful bowel movements
Fatigue
Activity limitation

Longitudinal Findings

Pelvic pain

Present across all three recorded cycles with highest frequency from approximately two days before menstruation through the first three days of bleeding.

Bowel symptoms

Painful bowel movements repeatedly clustered around menstruation and were substantially less frequent during the remainder of the cycle.

Functional impact

Nine days involved reduced normal activity and four days involved inability to complete an important activity.

Medication

Analgesics recorded on fourteen days. Severe symptoms persisted on five medication days.

Change

Non-menstrual pelvic-pain episodes became more frequent during the latest recorded cycle.

Clinical Timeline

Previous investigations, treatments and consultations appear chronologically.

Data Quality

79 of 92 days contained usable entries.

All information is explicitly identified as patient-reported unless another source is shown.

Every major statement is clickable back to its evidence.

---

44. Why the App Remains Usable for Years

The product should become **easier**, not harder, the longer somebody uses it.

```
WEEK 1
System learns the patient

MONTH 1
Frequently used symptoms become shortcuts

MONTH 2
Unnecessary questions disappear

MONTH 3
Personal baseline becomes meaningful

CONSULTATION
Longitudinal summary becomes valuable

TREATMENT CHANGE
System measures the new period

FOLLOW-UP
Previous history already exists
```

The application does not need gamification to create retention.

Accumulated longitudinal history is itself the value.

---

45. Reliability by Design

The architecture enforces:

```
No missing-data inference
No LLM calculations
No automatic diagnosis
No disease staging from symptoms
No unsupported report sentence
No hidden population risk score
No automatic clinician sharing
```

Every clinical-facing statement follows:

```
Patient event
      ↓
Deterministic calculation
      ↓
Derived feature
      ↓
AI explanation
      ↓
Evidence verification
      ↓
Patient approval
```

This makes the system auditable.

---

46. What We Deliberately Do Not Build Initially

No social network.

No advertising.

No symptom streaks.

No guilt-based engagement.

No “Endometriosis Risk: 87%.”

No automatic disease staging.

No permanent clinician access.

No mandatory sexual-health questions.

No mandatory fertility questions.

No uncontrolled medical web search.

No AI training on patient data by default.

No excessive wearable data without proven value.

---

47. MVP

The first complete product should contain:

```
Privacy-first onboarding
         ↓
Personalized symptom vocabulary
         ↓
30-second daily tracking
         ↓
Impact tracking
         ↓
Cycle engine
         ↓
Medication + treatment timeline
         ↓
Investigation + healthcare timeline
         ↓
Longitudinal pattern engine
         ↓
Evidence-grounded AI summary
         ↓
Patient review
         ↓
Clinician report
```

That already solves the fundamental problem.

48. Long-Term Expandability

The system is designed around generic observations rather than hard-coding today's endometriosis knowledge.

Future validated sources can therefore be added as new event types:

```
Wearable observation
Laboratory biomarker
Imaging-derived observation
Genomic result
New patient-reported measure
New diagnostic technology
```

without replacing the basic architecture.

The permanent model remains:

```
OBSERVATIONS
      ↓
LONGITUDINAL HISTORY
      ↓
DERIVED EVIDENCE
      ↓
INTERPRETATION
      ↓
COMMUNICATION
```

Clinical knowledge can evolve while historical patient observations remain unchanged.

---

49. The Key Product Decision

The important innovation is not how many symptoms the application can record.

It is that the system can support a **very rich patient model while exposing only the few questions relevant to that patient on that particular day**.

The complete underlying dataset can contain:

pain + location + character + timing + bowel symptoms + urinary symptoms + bleeding + fatigue + sexual symptoms + fertility context + treatment + medication + investigations + healthcare history + functional impact + patient-defined symptoms

without turning the daily experience into a questionnaire.

The product does this through:

progressive onboarding

conditional questions

personalized shortcuts

optional sensitive modules

voice-to-structured entry

event-based detailed logging

cycle-aware prompts

automatic feature calculation

logging by exception

and

patient-specific baselines

The patient supplies the experience.

The software handles the structure.

Final Design Principle

A successful version of this application should create an unusual outcome:

> **The dataset becomes richer while the patient's interaction becomes shorter.**

That is the central product-design challenge.

We do not ask the patient to behave like a researcher recording dozens of variables every day.

We design the system to quietly build a high-quality longitudinal record from very small interactions.

That gives the platform its final technical identity:

A privacy-first, adaptive longitudinal health-intelligence system that converts low-burden patient input into high-value, evidence-traceable clinical context.